Review of the factor-base line search (`research/jv_cover_fbline_20261102`, PR #3286)

This review has two parts, run independently on main at `5f3b5c209a`: a validator (`validator/REPORT.md`) and a red team (`red_team/REPORT.md`, with `red_team_report.yaml`). Neither read the other's report. Both harnesses blocked writing the report files themselves, so the main session wrote them from the text each returned, verbatim apart from a one-line header. The graph is `frontier.svg` and the PDF is `REVIEW.pdf`.

Bottom line

- **The measurement is sound and reproduces.** The line search cuts the Nagao-only cover arm's cost by **×1.71** [t(8): 1.50, 1.94; bootstrap: 1.54, 1.89], with the search charged.
  - The validator recomputed every number and replayed one run bit-for-bit.
  - The red team censused all ~10^6 lines per instance. The B^{-5} mechanism, the rate law and the per-test cost all hold.
  - The plain instances were not unlucky: the fair random-instance gain is ×1.71.
- **The framing "the cover passes the negation-map rho" is refuted.** The improved arm is not the frontier for this curve class. The repository's own sieving variant of the same cover (`jv_sieve.rs`, JV12 §3.2) costs S = 0.00615 at p = 1511 in the same unit. That is **83–100× cheaper** than the line-search arm (21–25× with its additions and lookups charged). It had already been measured at **0.0098 of the strong rho** in `bitop/v1` (`research/ic_acct1_cover_bitop_20261021`).
  - So the cover route was already far below rho.
  - #3286 is a component speed-up of a dominated arm, not a new crossing.
  - The comparison #3194, #3278 and #3286 made against rho should have been checked against that sieve frontier first. None of the three did.
- **U1 ("below rho in group additions") is downgraded.**
  - U1 prices a rho step at the harness's schoolbook 331 F_p multiplications, 2.40× the tuned 137.7. The cover gets no matching discount.
  - The cover's F_p additions are not emitted. With about one addition per multiplication (a model, not measured), the p = 1301 lead disappears once an addition costs 0.32 multiplications.
  - Under a strict §5a reading the arm cannot be ranked until those counters are emitted.
  - **U2 (F_p multiplications) is the like-for-like unit, and there the arm is above rho at all three primes:** 1.90 / 1.58 / 1.32.
- **Rho is the strongest available.** On all 9 instances $j(E) \notin F_{p^2} \cup F_{p^3}$ and $j \neq 0, 1728$, so $\mathrm{Aut}(E) = \{\pm 1\}$. The q-power Frobenius maps E to a non-isomorphic conjugate, so no √3 speed-up exists. (I had suggested one in the brief; that suggestion was wrong.)
- **The gain decays with p.** It goes as about $(1 + 4.2–4.3/\sqrt{p})^{-5}$ (model): ×1.57 at p = 2039, ×1.23 at $p = 10^4$. It is a constant-factor improvement, not an exponent change. With that decay in the slope, the U2 model crossover moves from $p^* \approx 1799$ to **≈ 1845–1865**, still inside u64.

Errata to the study README (recorded here; the merged study files are not edited)

1. **The sha** "**af4e1355…**" belongs to #3278's `negrho_rescore` tool. This study's own `fit/src/main.rs` is `f4149680`….
2. "**Controls (all pass)**" refers to C1–C4. The committed `analysis/rescore_fit.json` has `controls_pass: false`. These are #3278's consistency controls (they reproduce #3278's plain-arm numbers), and they fail by construction when the line records are the input. The README should have said so.
3. **The re-score command and layout were not committed.** The re-score fed the 9 line records (renamed `dlp_p*_s*.json`) as `inputs/cover/` of a copy of #3278's input tree, with `new/`, `old_rhostrong/`, `redteam_p251/` and `old_rhodp/` unchanged. With the plain records instead, the tool reproduced #3278's `table.md` byte-for-byte. Both the validator and the red team reproduced the numbers independently.
4. "**A tuned F4 is the remaining lever for U2**" is not supported. The sieve route, an exhaustive table-driven line search (+×1.075 measured ceiling) and other levers are untested.

Small uncharged items (validator, all negligible)

- The square table charges (p−1)/2 of the p−1 products it executes. About 700 multiplications per run go uncharged, a 3e-9 share.
- The search's 65,536·p table lookups are not emitted. Priced as native operations, they move U1 at w = 1 from 0.838 to 0.843. No verdict changes.

What this means for "closing the distance"

route (weak class, p ≈ 1500)	**S (U1 harness unit)**	**vs negation rho 0.93**
Nagao, plain line (#3194)	0.81–1.31	0.87–1.40
Nagao, best line (#3286)	0.51–0.74	0.55–0.79 (U1); 1.32–1.90 in U2
Sieve route (V2 T1, p = 1511)	**0.00615** (S^+ 0.0248)	about 0.0066–0.027; **0.0098 in** `bitop/v1` (ic_acct1)

The red team's pilot (exploratory, not frozen, not evidence) applied the study's `search_fb_line` to the sieve route at p = 1511, seeds 17–24, with all runs correct:

search size	**route only**	**search charged**
n = 65,536	×1.75	×0.92, a net loss (the search is 47% of the cost)
n = 4,096	×1.49	**×1.42 [1.18, 1.73]**

Recommended next step (red team, endorsed). Freeze a protocol applying the line search to `jv_sieve::run_cover_sieve_dlp_on`: - **Instances and options:** V2's (p ∈ {1009, 1511, 1823}, seeds 17–32, `--stop 64`, sparse). The plain arm must reproduce V2 exactly. - **Search cost:** charged in full. Fix n by a rule set before any data, or use a χ(h) table over F_q (about 38q multiplications plus priced native lookups). - **Pricing:** both arms in `bitop/v1` through the

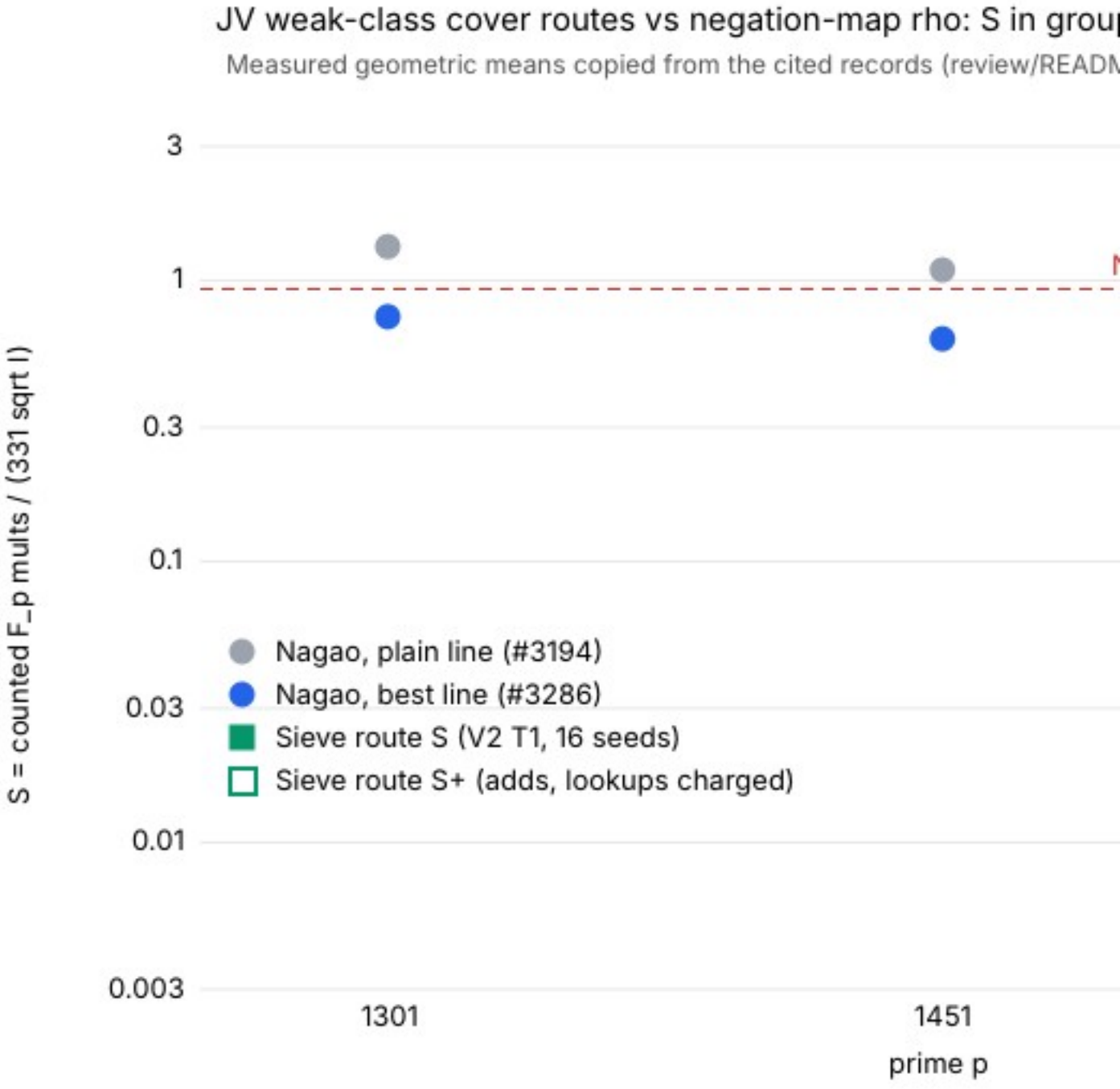
`jv_bitops` hooks, against the matched strong rho with the §5a band. - **Before any U1 ranking of the Nagao arm:** emit its F_p additions.

Validation gate

There is no `VALIDATION.json` for a leaderboard row. The validator passed the evidence as admissible for the paired component claim. The red team refuted the frontier framing. No leaderboard or `benchmarks/` row is warranted for the Nagao arm, because it is dominated.

Sources for the graph and table

- `research/jv_cover_crossover_20261101/results/dlp_*.json`
- `research/jv_cover_fbline_20261102/analysis/paired.txt`
- `research/jv_sieve_strong_rho_v2_20261020/TABLES.md`, T1 row p = 1511
- `research/jv_cover_negrho_rescore_20261101/table.md`
- `research/ic_acct1_cover_bitop_20261021/README.md`



Cover routes vs rho

Validator report: factor-base line search for the JV genus-3 cover (`research/jv_cover_fbline_20261102`)

Snapshot: main at `5f3b5c209acabf8ff948d95164b1508668e6b124` (PR #3286). The tree was clean apart from `review/`.

Validator: an independent session. It did not open any red-team report. Every number comes from its own Rust code (`check/`). Python was used only to edit that code's text, never for arithmetic.

Verdict: passed, with scope limits. The records are admissible evidence for the paired measured result on the 9 weak-class instances. They are not evidence of an asymptotic improvement. The U1 "below rho" ranking and the model p* values carry the limits in section 7.

0. Claim

`JV_FB_LINE=n` turns on a search over factor-base lines (`search_fb_line`, `line_poly`, `line_div`, `line_count`) and moves each instance to the isomorphic model H′ with the largest factor base. Claimed results: - Geometric mean of S_line/S_plain: 0.587 [0.530, 0.651], a 1.71× speed-up, over p ∈ {1301, 1451, 1601} × seeds 1–3. - Against #3278's negation-map rho, estimator (b) pooled (0.9322 in U1, 0.3879 in U2): - U1: 0.791 / 0.659 / 0.548, below rho. - U2: 1.900 / 1.585 / 1.318, above rho.

1. Receipts: pass

Checked field by field in `check_output.txt`, with 0 failures. - **Results.** 9/9 runs are solved and correct, with mismatches 0 and timed_out 0. One test was incomplete in arm B (p1301 s3) and one in arm A (p1451 s3); an incomplete test can only lose relations. - **Instance identity.** `l`, `c_add_e` (331) and `c_add_j` (2122) are equal between the arms on all 9. `stop_mode` is full in both arms. - `fb_line` **fields.** - candidates = 65536. - `b_plain` equals A's base: 634, 651, 662, 735, 729, 703, 764, 820, 824. - `b_best` equals B's base: 728, 725, 730, 811, 800, 804, 884, 883, 883. - best_index > 0 on all 9. - **Phase sums.** - total_muls = setup + base + stream + oracle + verify + la, exactly, on all 18 records. - oracle = weil + f4 + lin + post. - la_muls = 16·la_ops. - setup_muls ≥ fb_line.muls. - Setup minus the search differs from A by between −23,413 and +38,054. This comes from the stream start and step on the moved model, after the rng draws diverge. It is below 1e-7 of the total. - **Recomputation.** S = total_muls/(c_add_e·√l) matches the recorded value to 1e-12. decomposition_rate = relations/residuals. - **Search charge, derived independently.** (p−1)/2 + (17p+5) + (n−1)(17p+346) + 374. - It matches fb_line.muls on 6 of 9 runs. The other three are lower by 20, 10 and 10. - Each gap is a whole number of `pmul` zero-coefficient skips. Each skip is 2 F_q products, 10 multiplications not executed either. 4 skips were observed against about 6.0 expected. - **Search share:** 0.508–0.574% of each run. - **C4:** rate/((2B/p)^6/720) lies between 0.921 and 0.994, inside the band [0.8, 1.25]. - **C1:** every deterministic field is equal to the crossover record: residuals 428147, digest 4667961897523824477, total_muls 409975021213, decompositions 663, and all phases. Only

these differ: the timing fields, the racy multi-thread diagnostic `f4_muls_per_test_sum`, and the `fb_line: null` key, which the older schema lacks. - **C2:** - Residuals, digest and decompositions are equal to C1. - Total and setup are each higher by exactly 22,772 = fb_line.muls = (p−1)/2 + 17p + 5. - Nothing else differs.

2. Freeze ordering: pass

- **Commits.**
  - `PROTOCOL.md` was added in `0c2258ead2` (00:54:43Z).
  - `run.sh`, `FROZEN.sha` and `versions.txt` were added in `795eb00bed` (01:01:29Z).
  - `fit/` and `FIT.sha` were added in `112da2ea31` (01:03:03Z).
  - None of these was modified afterwards.
- **Hashes match.** PROTOCOL 96420747…, run.sh 1da32f05…, fit/src/main.rs f4149680…, Cargo.toml f273020c….
- **Arm B started between 01:20:18 and 01:24:55**, after both freezes. The lower bound is the binary's mtime (00:57) plus C1 (678 s) and C2 (720 s). The upper bound is the first arm-B artifact, which appears in `09b723b71f` at 01:24:55Z. The FIT.sha directory listing taken at 01:02:43Z contains no `line_*` file.
- **Binary.** The target/ binary before my rebuild is dabf56bc…, matching versions.txt.
- **Notes:**
  - C1's start is bounded only from above (≤ 01:01:39), so it may have begun seconds before the freeze commit. This is immaterial, because C1's pass criterion is fixed bit-equality.
  - walls.txt already existed at 01:02:43, consistent with the launcher creating it in advance. There is no evidence of a discarded attempt.
  - Arm-B wall time totalled 4,850 s, under the 4 h cap, so nothing was cut.

3. Independent recomputation: agrees to 3 dp

I used my own xoshiro256** bootstrap, type-7 quantiles and 20,000 resamples.

quantity	**claimed**	**validator**
S_B/S_A per instance	0.488 0.585 0.625 0.612 0.611 0.492 0.468 0.704 0.761	0.4878 0.5845 0.6252 0.6125 0.6108 0.4917 0.4679 0.7035 0.7609
geo-mean	0.5865 [0.5301, 0.6508]	**0.5865** [0.5295, 0.6495]
speed-up	1.705 [1.537, 1.886]	1.705 [1.540, 1.888]
model geo-mean $(B_{plain}/B_{best})^5$	0.5872	0.5872
spread of S across instances, plain → line	2.2× → 1.55×	2.245× → 1.550×
reference runs	126	126 admitted, 0 excluded, 14 per instance; `l` and `c_add_e` equal to the cover's; s_harness and s_tuned recompute to 2e-

quantity	claimed	validator
		16
pooled reference means	0.9322 / 0.3879	0.9322 / 0.3879
U1 at p = 1301 / 1451 / 1601	0.791 / 0.659 / 0.548	0.791 [0.720, 0.871] / 0.659 [0.607, 0.719] / 0.548 [0.506, 0.596], below
U2 at p = 1301 / 1451 / 1601	1.900 / 1.585 / 1.318	1.900 [1.731, 2.094] / 1.585 [1.459, 1.729] / 1.318 [1.215, 1.433], above
plain-arm “before” values	1.405 / 1.160 / 0.870; 3.376 / 2.787 / 2.090	same
§5a band (w = 0.1, 1)	rescore_table.md	same. U1 is below at every w. U2 is above at w = 0 and not resolved or below at w ≥ 0.1
model p*	1185 / 1799	1185 / 1799

- **H1 decision rule:** holds (upper end < 1 and every S_B < S_A).
- **Fit of the model:** log(measured) against log(model) gives r = 0.992, slope 1.20.

4. Mathematics and code

- **Isomorphism, not twist.**
  - The map $H' \to H$ is $(s, y') \mapsto (as+b, t^7 y')$ with $a = t^2$, $t \in F_q^\times$. Its inverse is $((x-b)/a, y/t^7)$.
  - $h = F \cdot N$ is monic of degree 7, so $h(as+b)/a^7$ is monic and the map is defined over F_q.
  - A non-square a would need $\sqrt{(a^7)} \notin F_q$, which gives the twist. The code only uses $a = t \cdot t$.
  - The single point at infinity is fixed, so the Jacobians are isomorphic over F_q and logarithms are preserved.
  - Remark: square a reaches only half of the $p+1$ line directions.
- **line_div.**
  - $u' = u(as+b)/a^{\deg u}$ is monic. $v' = v(as+b) \cdot t^{-7}$.
  - $v'^2 - h' = (v^2 - h)(as+b)/a^7$, which is divisible by u', and $\deg v' < \deg u'$.
  - The identity element is left unchanged.
- **Confirmed in the runs.** The `is_valid` assertion on G' and Q' held (rc 0 on all runs). The logarithm is checked on the original $E(F_{p^6})$ points, which the move does not touch.
- **Moved model used everywhere.** `Ctx::with_trace` moves `cov.hx`. These consumers all read the moved model:
  - `factor_base`
  - `weil_forms`
  - the y-recovery check
  - `verify_dec` (through `c.jac()`)
  - the residual stream
  - `unit_costs`
  - every worker context (`map_init` on the moved `spec`)

  `cov.nx`, `cov.fx`, `cov.d` and `transfer` are used only before the move or in other

experiments. `mitm6` is inactive (small > 40).

- **Charging, F_p multiplications in the harness unit.**
  - `line_count` charges 17p per line, which equals the `mm` calls it executes (Horner 14, norm 3).
  - The norm-based squareness test is valid: z is a square in F_q^× iff N(z) is a square in F_p^×, with F_q = F_p[i]/(i² − w).
  - line_poly, t², the inverse and the moves of G and Q go through the counted Fq operations (5 per product, 21 per inverse).
  - The total is read after the moves and enters setup_muls and total_muls.
- **Uncharged or unemitted items:**
  1. The square table executes p−1 products and charges (p−1)/2. That leaves 650–800 uncharged per run, about 3e-9 of the total.
  2. The search's 65536·p lookups (8.5e7–1.05e8 per run) are not emitted. As native operations, they move U1 at w=1 from 0.838/0.699/0.581 to 0.843/0.703/0.584, and U2 at w=0.1 from 1.050 to 1.054. No verdict changes.
  3. The search's F_p additions, about 15 per point per line, are not emitted (as for all cover additions).
  4. `line_poly` is recomputed in each Ctx (about 341 multiplications), re-deriving a value the run already holds. `Cover::new` and `is_valid` are uncharged on both arms.

5. Replay: bit-identical

- **Build.** `cargo build --release --example jv_cover` at main 5f3b5c209a (clean tree): 6 min 50 s, sha256 bb2b96d8….
  - It differs from the producer's binary because other modules changed, and `jv_cover.rs` had rustfmt-only edits since 0c2258ead2.
  - `rref_mod_p`, `square_core`, `wiedemann_u64` and `f4_fp` are unchanged.
- **Run.** `env -u JV_LIN JV_FB_LINE=65536 jv_cover --exp dlp --rho-runs 0 --rho-ref 1.3 --check-every 64 --sizes 1301 --first-seed 3 --seeds 1`: rc 0, 361 s on 4 cores.
- **Comparison.** 63 of 67 fields are identical:
  - total_muls 256314105656
  - residuals 265906
  - relation_digest 7945254227474834773
  - decompositions and relations 731, base 730
  - setup 1473116883, f4_muls 198197818058
  - f4_native_ops 13962232205, la_ops 13223299
  - solved and correct true
  - fb_line {b_plain 662, b_best 730, best_index 8085, muls 1472135851}

  Only wall_ms, f4_ms, lin_ms and f4_muls_per_test_sum differ. The log is identical apart from the wall time.

6. Heuristic, null-control and cost-model checks

- **Pre-registration.** The rate model, the B^{-5} scaling, the per-instance $(B_{plain}/B_{best})^5$,

C1–C4 and the decision rule were frozen before arm B. Predicted 0.5872; measured 0.5865. The largest per-instance deviation is +7.5% (p1601 s3: 0.761 against 0.708).

- **Null and dose response.**
  - C2 (n=1) is the null of the mechanism: the residuals and relations are identical, and the cost ratio is 1 + 5.6e-8.
  - The measured ratios follow the model with r = 0.992.
  - The plain bases' z-scores have sd 1.03, which confirms $\sigma_B \approx \sqrt{p}/2$. The best bases' z-scores average k = 4.20, the expected maximum of 65536 draws.
- **Decay.** At fixed n the gain fades with p: $g(p) \approx (1 + 4.2/\sqrt{p})^{-5}$, which is 0.577 at p=1301, 0.641 at 2039, 0.728 at 4096 and 0.922 at 65536. The README does not state this (L1).
- **Unit.** The unit is declared: harness F_p multiplications / ($c_{add_e} \cdot \sqrt{l}$). U2 compares harness counts on the cover with tuned counts on rho; this is stated and inherited from #3278. Memory is not reported for either arm, and byte counters are not charged under §5a.
- **§6 ladder.** The primary claim is measured as a paired ratio on the stated instances. Only p* is a model, and it is labelled as one. The reproducibility pointer was exercised (section 5).

7. Limitations and scope

- **L1.** This is not an asymptotic improvement. At fixed n the gain is about $(1 + 4.2/\sqrt{p})^{-5}$, and the 1.71× applies only to p ≈ 1301–1601.
- **L2.** The model p* reuses the plain arm's slope (−2.10) and ignores L1. Adding the gain's p-dependence ($d \ln g / d \ln p \approx +0.24$) moves U2 p* from 1799 to about **1850** and U1 p* from 1185 to about 1152. Both remain model values, both ≤ 2039.
- **L3.** "U1 below at every prime" ranks a cover whose cost is a lower bound: its F_p additions are not emitted, while rho's are (0.127 S-units).
  - Assume one addition per counted multiplication (not measured). Then the U1 point estimate reaches 1 when an addition costs α^* = **0.32** multiplications at p=1301, 0.65 at p=1451 and 1.10 at p=1601.
  - With 0.5 additions per multiplication, α^* is 0.81, 1.76 and 3.28.
  - So the 1301 "below" is fragile under complete counts. `research/ic_acct1_cover_bitop_20261021` could settle the additions-per-multiplication ratio.
  - The paired S_B/S_A is unaffected.
- **L4.** There are 3 seeds per prime and one residual stream per instance. The rng draws diverge after `factor_base`, so the pairing is per instance.
- **L5.**
  - The runs were local, not on isolab-eks; this is disclosed.
  - How the line-search records were staged into negrho_rescore's `inputs/cover/` input is not recorded. My numbers reproduce it.
  - The README's "fit/src/main.rs sha af4e1355…" refers to #3278's tool, not this study's `fit/` (f4149680…).
- **L6.** The claim covers the weak class only (full rational 2-torsion, #E = 4l), with $l \approx 2^{60–2}62$.

- **L7.** The square-table undercharge and the unemitted search lookups and additions are quantified in section 4 and are negligible.

8. Files

- `check/` (Cargo.toml, src/main.rs, Cargo.lock). Build and run with `CARGO_TARGET_DIR=<tmp> cargo build --release --offline && <tmp>/release/fbline_validator_check <repo> <out> [replay.json]`.
- `check_output.txt`.
- `replay/` (REPLAY_ENV.txt, line_p1301_s3.json, line_p1301_s3.log).
- `artifact_hashes.txt` and `reference_runs.sha256`.
- `validation_report.yaml`.

Red team: factor-base line search for the JV cover (`research/jv_cover_fbline_20261102`)

Report id `RT-20261102-172964`. The suffix is a random 6-hex value, because `allocate_id.py` has no RT type.

Snapshot reviewed: `main` at `5f3b5c209a`, which contains PR #3286. `src/cryptanalysis/jv_cover.rs` differs from the study's build commit `0c2258ead2` only by rustfmt line breaks. The review was read-only. I did not read the sibling validator report; I saw only its commit subject.

Labels: **measured** means read from a run record. **Derived** means computed in Rust from records. **Model** means an extrapolation or assumption, named as such. **Exploratory** means a red-team pilot that is not frozen and is not evidence.

Provenance: - My binaries link the compiled `crypto_lib` at main: rlib sha256 `c8a3a56c…6e41`, rustc 1.97.0. - `line_census` sha256 `e91c947b…09cf`. - `sieve_transfer_pilot` sha256 `ded3fc4d…2724`. - `fbline_rt_records` sha256 `6afb3178…cd60`.

Verdict per claim

#	Claim	Verdict	Numbers
1	1.71× on the 9 instances	survives	Geo-mean 0.5865 measured, 0.5872 model. t(8): ×1.705 [1.499, 1.939]; the bootstrap gave [1.54, 1.89].
2	Not regression to the mean	survives	Plain-line percentiles 0.03–0.88 (mean 0.507). Mean z(B_plain) −0.02. Fair speed-up ×1.710

#	Claim	Verdict	Numbers
			against ×1.705 realised.
3	Rate ∝ $(2B/p)^6$; per-test cost unchanged	survives	Paired log(rate/model), line − plain: +0.005, t(8) = 0.45. F4 per residual ×1.0001; linear stage ×1.0011.
4	Search fully charged	survives	0.51–0.57% of each run. Native lookups 0.05–0.23 GAE/√ℓ; §5a band unchanged.
5	U1 below rho (0.791 / 0.659 / 0.548)	survives arithmetically; downgraded in meaning	Rho is the full-automorphism rho. t(2) still below. But a rho step is priced at the schoolbook 331, 2.40× the tuned 137.7, and §5a strict admissibility fails (unemitted cover counters).
6	U2 above (1.900 / 1.585 / 1.318)	survives; like-for-like unit	τ* = 0.526 / 0.631 / 0.759.
7	Rho reference matched and strongest	survives	Same ℓ and c_add_e on 126 runs; `--neg 1 --r 256 --batch 16`. $j \notin F_{p^2} \cup F_{p^3}$, $j \neq 0$, 1728 on all 9 instances. No Frobenius speed-up, no √3.
8	Framing “passes the negation rho”	refuted as framing	The in-tree sieve variant has S = 0.00615 at p = 1511. The line arm costs 83–100× more (21–25× more than the sieve’s S^+).
9	p* at slope −2.10	weakened	d ln G / d ln p = −0.26, so the slope is about −1.84. U2 p* ≈ 1845–1865 (not 1799); U1 p* ≈ 1145–1157.
10	Statistics	survives with a caveat	t-intervals wider,

#	Claim	Verdict	Numbers
			verdicts unchanged. sd of log(measured/model) 0.035, below the 0.054 count-noise floor.
11	“Controls (all pass)”	minor objection	`analysis/rescore_fit.json` has `controls_pass: false` (“consistency controls … FAILED”), expected on line inputs but not disclosed. The byte-for-byte reproduction and the re-score command are not recorded.
12	“Tuned F4 is the remaining lever”	under-supported closure	Untested: sieve route, large primes (`ic_cover_leads` L6), an exhaustive table line search (+×1.075), other F_p-sublines.

The strongest objection: the improved arm is not the frontier for this class

What the study improves. It speeds up `run_cover_dlp_mode`, which finds relations with Nagao six-point tests only.

The repository’s sieving variant of the same route. `jv_sieve.rs` implements [JV12] §3.2 with: - the same `generate_spec` class, - the same F_p-abscissa factor base (mean B = 756 at p = 1511), - the same Jacobian and the same six-point test for the descent.

Same unit, measured (V2 T1, 16 seeds). The sieve route has S = 0.00615 [0.0055, 0.0069] at p = 1511 and S^+ = 0.0248. The line-search arm has S = 0.615 at p = 1451 and 0.511 at p = 1601:

comparison	at p = 1451	at p = 1601
line-search arm S / sieve S	×100	×83
line-search arm S / sieve S^+	×25	×21

Pareto. Both routes have the same factor base and relation count, so memory is comparable. The line-search arm is dominated.

Bit operations. `research/ic_acct1_cover_bitop_20261021` measured the sieve route’s whole method at 0.0098 [0.0069, 0.0144] of the strong rho at p = 1511.

These numbers are current. My pilot reproduced V2's plain `total_muls` on 8 of 8 seeds at current main. Leaderboard Table D withholds only the sieve route's *rho* comparison; this comparison is cover against cover and needs no rho.

Exploratory pilot. The study's unchanged `search_fb_line` placed in front of `run_cover_sieve_dlp_on`, at p = 1511, seeds 17–24, all runs correct:

search size	route only	search charged	search share of the line arm
n = 65,536	0.571 [0.490, 0.664] (×1.75)	1.082 [0.930, 1.259] (×0.92)	47%
n = 4,096	0.670 [0.554, 0.812] (×1.49)	0.702 [0.579, 0.850] (×1.42)	4.5%

Hidden assumptions and the random model (claims 2 and 3)

Census of all lines. I computed B for every square-direction line $a \cdot F_p + b$: 0.85–1.28 M lines per instance. - **Replication.** Every `b_plain`, `b_best` and `best_index` matched the records, using the study's own RNG stream, on 9 of 9 instances. - **Distribution.** Mean ≈ p/2 (650.50 / 725.50 / 800.50). sd = √p/2 to within 0.2%. Skew within 0.004 of zero, excess kurtosis within 0.009. The Gaussian random-line model holds where it is used.

Extreme values. - z(B_best) averages 4.22 ± 0.18; extreme-value theory predicts 4.29 ± 0.28. - The exact census expectation of the maximum of 65,536 lines at p = 1301 is 727.3 / 726.6 / 727.0, against observed 728 / 725 / 730. - P(max ≥ observed) lies in 0.27–0.96.

Were the plain instances unlucky? No. - Percentiles of the plain line: 0.18, 0.51, 0.74, 0.69, 0.57, 0.12, 0.03, 0.84, 0.88. - Their mean, 0.507, is consistent with uniform (0.5 ± 0.096).

Fair speed-up. E[residuals for the plain line] / E[residuals for the best of 65,536], search charged: - ×1.710 overall (geo-mean); - ×1.757 at p = 1301, ×1.709 at 1451, ×1.666 at 1601.

What the quantity should do as the destroying parameter grows. The gain comes from $\sigma_B/\mu_B = 1/\sqrt{p}$, so it should decay like $(1 + 4.29/\sqrt{p})^5$ (model): ×1.57 at p = 2039, ×1.23 at p = 10^4, ×1.07 at p = 10^5. The measured per-prime ratios 0.563 / 0.569 / 0.630 decay as predicted. A random-line null would give a ratio of about 1 (model), since the plain line is a typical line. C2 already passed as the machinery null.

Rate law and per-residual cost. - Rate against the falling-factorial model $64 \cdot C(B,6)/p^6$ (geo-means): 0.981 plain, 0.987 line. Choosing the max-B line carries no rate penalty. - F4 multiplications per residual: 745.3–745.6 k on both arms. - Linear stage: 205–206 k on both arms. - Weil, stream and post work: +2–4% (more decompositions per residual; under 0.6% of the total). - The sum of phase counters equals `total_muls` exactly on all 18 records.

Cost model (claim 4)

What is charged. - Per line: $p(2 \cdot 7 + 3)$ multiplications. - Once: $(p-1)/2$ for the square table. - 21

per `line_poly` inversion. - The `line_div` moves of G and Q. - Total $1.47–1.81 \cdot 10^9$ per run, consistent with C2's 22,772 for a single line ($17p + (p-1)/2$ plus a few).

What is not charged, and does it matter. - Native: $65{,}536 \cdot p$ table lookups, 0.05–0.23 GAE/$\sqrt{\ell}$ for 1–3 native operations per point. U1 stays below in every band cell; worst case 0.852 / 0.710 / 0.590 at $w = 1$. - Point counting in `generate_spec`: instance generation, uncharged on both arms.

The search is overpriced (a candidate, not evidence). - A $\chi(h)$ table over F_q costs about $38q$ multiplications ($8.7 \cdot 10^7$ at $p = 1511$), 20× less than the study's search. - After that, each line costs only additions and lookups. - The exhaustive census ran in seconds per instance on 4 cores. - The global maximum adds a further ×1.075 (×1.025–×1.111). - Its $p^3/2$ lookups must then be priced in the §5a band.

Units (claims 5 and 6)

U1 is unfair to rho by up to 2.40×. - U1 converts the cover's work at the harness price of a rho step: 331 F_p multiplications (schoolbook F_{q^3} product 55, per-step inversion 166). - The tuned step costs 137.7 ($s_{tuned}/s_{harness} = 0.4161$). - The cover's work is 77% F4 and 21% linear stage, all native F_p, with no extension-field arithmetic to discount.

U2 is close to like-for-like. - `f4_fp` matches msolve wall time on generic benchmark systems (#1717). - The linear stage is already the sparse/fast variant. - `--trace` replay is a measured regression (`ic_cover_leads` L3), and `--stop` staircase saves nothing (L4). - What remains is unmeasured: F5 criteria (≈1.08×, heuristic) and an early F4 stop (at most 37% of F4 rows, speculative).

Tuned-cover model (labelled model): tuning factor $\tau_c \in [0.65, 1.0]$, so tuned cover against tuned rho is $U2 \cdot \tau_c$.

p	tuned / tuned	reading
1301	1.24–1.90	above
1451	1.03–1.58	above to parity
1601	0.86–1.32	around parity

F_p additions (model: the cover emits none and makes about $r_c \approx 1$ addition per multiplication; rho emits 45.2 per step).

- **Physical price, 0.1–0.3 of a multiplication:**
- **Additions priced as §5a native operations:** U1 flips to above at $w = 0.1$ (3.94 / 3.28 / 2.73).
- **Admissibility:** §5a says an arm with an unemitted named counter "cannot be ranked", yet the README declares "the order holds".

Cheapest discriminating control. Run one line-search instance per prime with the `jv_bitops bitop/v1` tally on. The tooling exists (`examples/jv_cover_ic_vs_rho_bitops.rs`).

Rho reference (claim 7)

Matched: - `l` and `c_add_e` equal on all 126 runs. - Flags `--neg 1 --batch 16 --r 256 --dp-bits 16`. - Positive control z = +1.15.

Automorphisms (measured): - On all 9 instances, j(E) is in neither F_{p^2} nor F_{p^3}, and $j \notin \{0, 1728\}$. - The q-power Frobenius maps E to E^σ, which is not isomorphic to E because $j^q \neq j$. It is not an endomorphism of E, so there is no order-3 class speed-up. - There is no GLS map ψ with $\psi^2 = -1$, because $j \notin F_{p^3}$. - So Aut(E) = {±1}, and the negation map is the full-automorphism rho.

A hypothetical √3 would have given U1 = 1.37 / 1.14 / 0.95; it does not exist. Rho slack (model): about 15% at most from a larger inversion batch, which would only widen U2 against the cover.

Statistics (claim 10)

- **Paired ratio:** the t(8) interval 0.5865 [0.5156, 0.6671] is wider than the bootstrap's [0.5301, 0.6508]. The conclusion is unchanged.
- **H1's "every instance cheaper" is almost guaranteed by design:** the plain line is candidate 0, so B_best ≥ B_plain. The informative test is the model match.
- **Per prime, t(2) on the cover side combined with the reference standard error:**

Crossover (claim 9)

The README applies the plain arm's slope, −2.10, to the line arm. The line-search gain decays with p (d ln G / d ln p = −0.26), so the line arm's slope is about −1.84 (model). Anchored at each measured prime: - **U2 p*: 1844 / 1864 / 1860, against 1766–1826 at slope −2.10 and the README's 1799. - U1 p*:** 1145–1157.

The planned run at p ≈ 1850–2000 still fits inside the u64 limit.

Narrowest supported statement

Setting. On the 9 `jv_cover` weak-class instances (p = 1301, 1451, 1601; genus-3 cover; #E = 4ℓ; never prime order), choose the best of 65,536 square-direction F_p-lines, with the search charged.

Effect. This cuts the cost of the Nagao-only arm by ×1.71 [1.50, 1.94]. - The cut matches the B^{-5} model; the rate law and the per-test cost are unchanged. - The fair expected gain for a random instance is ×1.71. - It decays as $(1 + 4.29/\sqrt{p})^5$ (model).

Against the matched negation rho, which is the full-automorphism rho: - In like-for-like F_p multiplications (U2), the arm is **above**: 1.90 / 1.58 / 1.32. - In harness schoolbook-priced group additions (U1), it is **below**: 0.79 / 0.66 / 0.55. That is a lower-bound comparison, because the cover's additions and pair selection are not emitted.

Frontier. The arm is 83–100× costlier than the in-tree sieve variant of the same cover route. Until the lever is shown on that route, this is a component improvement of a dominated arm.

Next concrete action

What to freeze. A protocol applying the line search to `jv_sieve::run_cover_sieve_dlp_on`.

Search cost. Charge it in full, and either: - use the χ(h) table (about 38q multiplications plus n·p native lookups, priced in the band), or - fix n by a rule set before any data.

The pilot shows n = 65,536 is a net loss there, and n = 4,096 gives ×1.42 [1.18, 1.73].

Instances and options. V2's: p ∈ {1009, 1511, 1823}, seeds 17–32, `--stop 64`, sparse. The plain arm must reproduce V2 exactly.

Pricing. Both arms in `bitop/v1` through the existing `jv_bitops` hooks, against the matched strong rho with the §5a band.

Before any U1 ranking of the Nagao arm. Emit its F_p additions.